

DONGJUN KIM

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Seoul, South Korea



AI Research Engineer on Upstage's Solar Model Project, building the evaluation systems behind frontier LLMs.
Specializing in AI safety through mechanistic interpretability and model evaluation,
with first-author oral papers at ACL 2026 and EMNLP 2025.

EXPERIENCE

AI Research Engineer

Mar 2026 – Present

Upstage, Evaluation Team (Solar Model Project)

Seoul, South Korea

- Run end-to-end evaluation of **Solar Open 2** (Korea's Independent AI Foundation Model project) and **Solar Pro 4**: one-command campaigns over a **50+ benchmark suite** on multi-node GPU clusters at every training stage, directly informing training and data decisions.
- Designed **light benchmark variants** for rapid checkpoint screening, cutting core-suite wall-time by **86%** (42 to 6 minutes) with score fidelity validated through repeated-run variance analysis and per-category sampling.
- Migrated and hardened **agentic benchmarks** (SWE-bench Verified/Pro/Multilingual, GAIA, MCP-Atlas, Terminal-Bench, τ^2 -bench, CritPT) with reproduction studies against reported scores; fixed timeout semantics, grading environments, and reasoning-output parsing.
- Reduced evaluation wall-time through systems work: tail-latency analysis, multi-completion sampling (up to **1.65×** on repeat-sampling math benchmarks), prefix-caching studies, and a deterministic scheduler that auto-tunes benchmark ordering and parallelism under cluster-throughput and sandbox-concurrency constraints.
- Built automated **leaderboard reporting**: incremental per-benchmark upload with pass@k, response/reasoning length, truncation-ratio, and degeneration monitoring.
- Aligned in-house results with external indices (e.g., *Artificial Analysis*), root-causing score gaps to judge choice, prompts, temperature, and a streaming reasoning-parser bug.

NLP & AI Researcher

Jan 2024 – Feb 2026

NLP&AI Lab, Korea University

Seoul, South Korea

- **Research**: Co-authored **10+ papers** at top-tier venues (ACL, EMNLP, NAACL), including two first-author main-conference oral presentations.
- **Evaluation**: Drove benchmark design and benchmark-profiling analyses; established reusable evaluation recipes, regression policies, and dataset/prompt versioning used across lab and consortium model iterations (incl. VAETKI for the WBL national project).
- **LLM Training**: Supported Korean LLM post-training and reasoning improvements, including **GRPO in VERL** for KULLM-R and instruction tuning plus data curation for KULLM3 and Ko-Gemma.
- **Mechanistic Interpretability**: Developed **SAE-based** feature discovery, feature steering, and causal-intervention workflows to diagnose behaviors, quantify feature effects, and connect internal signals to evaluation outcomes.
- **Inference Optimization**: Improved serving and evaluation throughput with vLLM and cache-aware decoding settings, balancing latency, determinism, and large-batch benchmarking stability.
- **MLOps & GPU Infrastructure**: Operated shared infrastructure (**40+ A100/H100 GPUs**) for 30+ researchers: Kubernetes, Docker environments, experiment tracking, artifact hygiene, and reproducible runbooks.
- **Safety**: Built interactive evaluation demos and safety-audit harnesses: jailbreak and refusal checks, PII and toxicity screening, and regression monitoring for risk-sensitive deployments.

RLHF Data Trainer

Mar 2023 – Jan 2024

Scale AI

San Francisco, CA (Remote)

- Generated high-difficulty **preference and instruction data** for alignment workflows (PPO, DPO), with emphasis on **coding and mathematics** and other reasoning-heavy domains.
- Performed **QA and peer review** to maintain rubric consistency and data reliability across trainers, delivering structured feedback through client tooling (e.g., OpenAI Feather).
- Collaborated directly with client engineers to identify emerging alignment gaps and rapidly translate requirements into updated production guidelines and examples.
- Covered multi-modal and safety-sensitive scenarios (image reasoning, multi-turn conversations, PII, harmful content) to support robust fine-tuning and evaluation.

AR/VR Software Engineering Intern

May 2023 – Nov 2023

Simacro

Cambridge, MA

- Developed Unity-based VR/AR applications transforming static P&IDs into interactive digital twins, streamlining operations for industrial clients including Hyundai Oil Bank.
- Built a computer-vision API (Python/C#) for symbol detection, automating **80%** of manual labeling at **95%** accuracy and accelerating the P&ID digitization workflow.
- Integrated Virnect's image-tracking SDK into Unity apps, enabling robust, anchored AR overlays of diagrams onto physical industrial machinery.

High Performance Computing Intern

Aug 2022 – May 2023

Dr. Edwin Michael's Lab, University of South Florida

Tampa, FL

- Built the **CitySEIRCast** digital-twin platform for city-scale pandemic forecasting, processing large datasets with parallel and distributed computing (MPI, OpenMP, CUDA).
- Developed data pipelines in Python (NumPy, Pandas, SQL) to manage simulation I/O; optimized C++ and Python simulation code for HPC clusters.

Mixed Reality Research Assistant

Jan 2022 – May 2023

USF Mixed Reality Lab

Tampa, FL

- Designed an automatic room-mapping method for Mixed Reality, reducing manual mapping time by **40%**.
- Published research on novel data collection and modeling techniques in MR at IPMV 2023.

EDUCATION

Korea University

Seoul, South Korea

Master of Science in Computer Science

Feb 2026

- Advisor: Prof. Heuiseok Lim (NLP&AI Lab)
- Research focus: LLM evaluation, mechanistic interpretability, and AI safety

University of South Florida

Tampa, FL

Bachelor of Science in Computer Science (Cum Laude)

May 2023

PUBLICATIONS

LANGSAE EDITING: Improving Multilingual Information Retrieval via Post-hoc Language Identity Removal

Kim, D., Yoon, J., Park, C., & Lim, H.

Proceedings of ACL 2026 (Oral Presentation)

Exploring Coding Spot: Understanding Parametric Contributions to LLM Coding Performance

Kim, D., Kim, M., Chun, Y. C., Park, C., & Lim, H.

Findings of ACL 2026

MMAC: A Multilingual, Multimodal Alignment Framework for Cultural Grounding Evaluation

Zheng, W., ..., **Kim, D.**, ..., & Chen, N. F.

Proceedings of ACL 2026

CrossDocVQA: A Benchmark for Multi-Hop Cross-Document Visual Question Answering

Shim, G., Shin, J., **Kim, D.**, Park, C., & Lim, H.

arXiv preprint, 2026

Benchmark Profiling: Mechanistic Diagnosis of LLM Benchmarks

Kim, D., Shim, G., Chun, Y. C., Kim, M., Park, C., & Lim, H.

Proceedings of EMNLP 2025 (Oral Presentation)

KoLEG: On-the-Fly Korean Legal Knowledge Editing with Continuous Retrieval

Seo, J., Jung, D., Lee, J., Chun, Y. C., **Kim, D.**, Ryu, H., Shin, D., & Lim, H.

Findings of EMNLP 2025

Enhancing Automatic Term Extraction with Large Language Models via Syntactic Retrieval

Chun, Y. C., Kim, M., **Kim, D.**, Park, C., & Lim, H.

Findings of ACL 2025

Towards Computational Comprehension: A Non-Anthropocentric Framework for Evaluating LLM Understanding

Kim, D., Shim, G., Kim, M., Park, C., & Lim, H.

arXiv preprint, 2025

From Snapshot to Stream: A Self-Improving Leaderboard for Robust and Evolving NLP Evaluation

Park, C., Moon, H., Kim, D., Lee, S., Seo, J., Eo, S., & Lim, H.

arXiv preprint, 2025

KITE: A Benchmark for Evaluating Korean Instruction-Following Abilities in Large Language Models

Kim, D., Park, C., Park, C., & Lim, H.

arXiv preprint, 2025 (under review at AACL 2026)

Exploring Inherent Biases in LLMs within Korean Social Context

Lee, S., Kim, D., Jung, D., Park, C., & Lim, H.

NAACL 2024 Student Research Workshop

CitySEIRCast: An Agent-Based City Digital Twin for Pandemic Analysis

Bilal, S., ..., Kim, D., ..., & Michael, E.

Complex & Intelligent Systems, 2024

PROJECTS

VAETKI: Independent AI Foundation Model Project (WBL)

Collaboration with NC AI and ETRI

- Owned engineering of the large-scale evaluation pipeline, standardizing **50+ benchmarks** into a single reproducible runtime with automated regression tracking across checkpoints.
- Implemented end-to-end checkpoint-to-evaluation automation, triggering evaluations immediately after checkpoint creation to shorten the training feedback loop.
- Integrated **vLLM** to accelerate inference throughput and improve evaluation turnaround time at scale.
- Built **W&B** dashboards and **W&B Weave** trace analysis to debug reasoning failures, visualize inference traces, and support slice-level comparisons.
- Implemented **data-mixture contribution analysis** to quantify which dataset combinations drove metric gains, translating results into actionable data and recipe updates.
- Introduced contamination defenses and runbooks, including deduplication and overlap scans, to support fair and comparable evaluations.

KULLM-R: Korean Reasoning Model Training nobrand/KULLM-R

NLP&AI Lab, Korea University

- Implemented **GRPO** reinforcement learning in **VERL** with multi-rollout group scoring, enabling critic-free policy optimization for Korean reasoning tasks.
- Designed **verifiable custom reward functions** optimizing correctness, Korean final-answer consistency, and an **adaptive length penalty** that reduces verbosity on easy problems while preserving depth on hard ones.
- Tuned reward weights and RL hyperparameters (KL coefficient, rollout settings, max response length) to balance accuracy, compute cost, and response quality.
- Ran iterative evaluations on Korean math and reasoning benchmarks (Pass@1 and length diagnostics), using results to refine reward shaping and training stability.

KULLM3 & Ko-Gemma Model Training nlpai-lab/KULLM3, nlpai-lab/ko-gemma-7b-v1NLP&AI Lab, Korea University

- Contributed to **post-training** in a 10-person team, including instruction tuning, training-framework improvements, and multilingual plus code-switch dataset development.
- Curated coding and math corpora, established quality gates, and ran capability evaluations for coding and mathematics.

Self-Improving Leaderboard

NLP&AI Lab, Korea University

- Implemented daily crawlers across multiple news categories, **real-time QA generation**, and **automated multi-LLM evaluation** on daily refreshed data.
- Launched a live leaderboard with time-aware ranking and quarterly stability and volatility metrics to track consistency over time.
- Maintained scheduling, monitoring, and data hygiene to support regular refreshes and clear longitudinal comparisons.

TECHNICAL SKILLS

Core ML & Deep Learning: PyTorch, JAX, Hugging Face (Transformers, Accelerate), Polars

LLM Training & Scaling: DeepSpeed, FSDP, Megatron-LM, PEFT (LoRA, QLoRA), SFT

RL & Evaluation: GRPO, GSPO, RLHF, RLVR (VERL, TRL), LM-Eval-Harness, EvalChem, LLM-as-a-Judge

Inference Optimization: vLLM, SGLang, TensorRT-LLM, KV-cache optimization, quantization (AWQ)

Infrastructure & MLOps: CUDA, Triton, Docker, Kubernetes, Slurm, Terraform, AWS (SageMaker), W&B